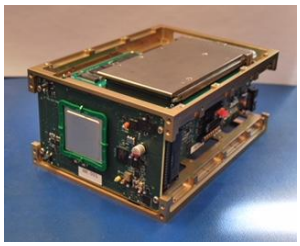
TechPort

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Terabyte Infrared Delivery 200Gb/sec Laser Communications System for LEO Direct-to-Earth Missions

Program: Space Communications and Navigation **Acronym:** TBIRD **Project ID:** 95868

Project Description



Satellites in low-Earth orbit (LEO) have on-board sensors that can generate large amounts of data to be delivered to a ground user. Direct-to-Earth delivery from LEO is challenging because of the sparse contact with a ground terminal, but the short link distances involved can enable very high data rates by exploiting the abundance of spectrum available at optical frequencies. NASA's Terabyte Infrared Delivery (TBIRD) program will demonstrate a direct-to-Earth laser communication link from a small satellite platform to a small ground terminal at burst rates up to 200 Gbps. Such a link is capable of transferring several terabytes per day to a single ground terminal. The high burst rates are achieved by leveraging off-the-shelf fiber-telecommunications transceivers for use in space applications. A 2U TBIRD payload is currently being developed for flight on a 6U NASA CubeSat. The relatively short distance (~ 1000 km) and corresponding reduced range loss of a direct-to-Earth (DTE) LEO laser link offers an opportunity to provide extremely high bandwidth downlinks at very low cost, size, weight, and power (SWaP). Under NASA SCaN support, MIT Lincoln Laboratory is developing the TeraByte InfraRed Delivery (TBIRD) terminal [6,7] to demonstrate a 200 Gbps downlink using only 1.8U x 1U x 1U volume (1U=10 cm) and less than 2.25 kg. The two baseline 100 Gbps modems are variants of a highly-integrated COTS design that was modified to be space-qualified; current efforts are now focused on the on-board 2.0 terabyte (TB) memory and controller which must be read at similar data rates. A demonstration flight is planned on the NASA Space Technology Mission Directorate (STMD) Pathfinder 3 CubeSat in late 2020 and it is expected to demonstrate downloads of the entire 2.0 TB data buffer on a single pass. While the initial demonstration will use existing NASA optical ground stations with adaptive optics (AO), there are plans to develop low-cost optical ground stations to support such rates to be specifically co-located at mission data storage facilities or even commercial data centres to reduce the cost of ground-based fiber backhaul at such data rates.

Anticipated Benefits

Low-Earth orbit is a desirable regime for many remote sensing and Earth-observing satellites due to its close proximity to Earth. Traditionally, these missions have been accomplished with large satellites (>500 kg) that often carry multiple exquisite instruments which may take measurements on multiple frequency channels [1]. The cumulative data volume that such instruments generate can be quite large as well. For example, the Terra satellite (launched in 1999 and still operating) carries a multispectral sensor suite that produces nearly 200GB of data per day onboard the spacecraft [1]. Single-instrument satellites can also generate hefty data volumes. For example, ICESat-2, launched in 2018 and now operational, carries a lidar instrument that produces up to 70GB per day after significant on-board compression [2]. Some upcoming science missions are targeting much larger data volumes. For example, the NISAR satellite (expected launch 2021) performs synthetic radar imaging (SAR) and plans to collect at least 4300GB (or 4:3 TB) per day for delivery to a ground network [3]. As another example, the medium-sized 200 kg satellite for the SWOT mission expects to generate 0:9 TB of daily with its radar instrument after 20x compression [4].

Organizational Responsibility

Organizational Responsibility	Public
Responsible Mission Directorate	Space Operations Mission Directorate (SOMD)
Responsible Program	Space Communications and Navigation (SCaN)
Lead Organization	NASA Headquarters (HQ)

Project Duration

Start: 2017-09-01 ⟶ End: 2021-12-31

Partner Organizations

Organization Name	Role	Type
NASA Headquarters	Lead Organization	NASA Center
Ames Research Center	Supporting Organization	NASA Center
Goddard Space Flight Center	Supporting Organization	NASA Center
Massachusetts Institute of Technology Lincoln Laboratory	Supporting Organization	FFRDC/UARC

Project Contacts

Contact Name	Contact Role
Beth A Keer	Project Manager
Bryan Robinson	Principal Investigator
Don Boroson	Principal Investigator
Kendall D Mauldin	Co-Investigator
Kevin M Coggins	Program Director
Gregory W Heckler	Program Manager

Primary U.S. Work Locations

California | District of Columbia | Maryland | Massachusetts

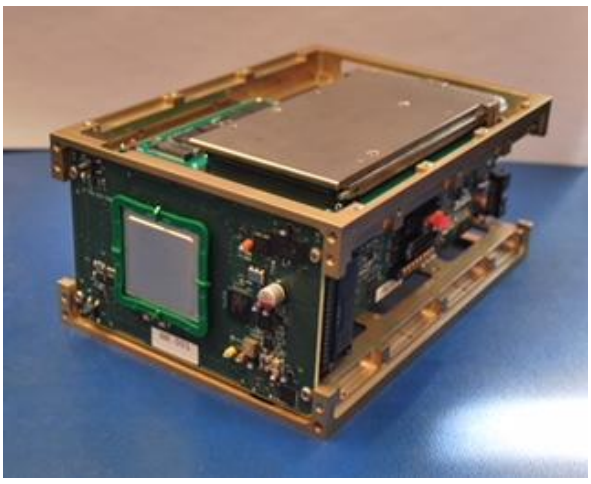
Technology Area

Technology Maturity

Start	2
Current	5
End	9



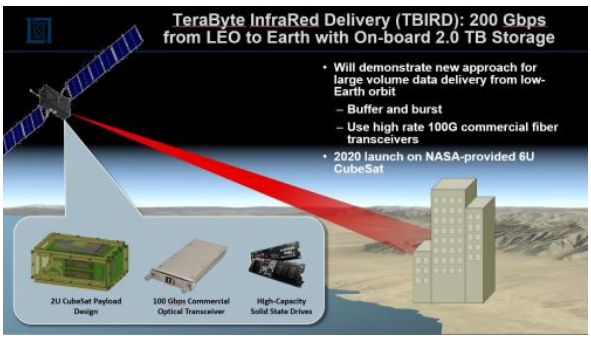
Project Library Images



TBIRD 200 Gbps Laser Communications Engineering Model

https://techport.nasa.gov/api/file/40249

TBIRD 200 Gbps Laser Communications Engineering Model with integrated 2.0 TB Buffer (2.3 kg, 1.9U volume)



TBIRD Concept of Operations for 200 Gbps Laser Downlink

https://techport.nasa.gov/api/file/40250

TBIRD Concept of Operations (OV-1) for 200 Gbps Laser Downlink


Earth